

Web Data Integration Project Abstract (Group 6)

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Abstract. We assess the feasibility of integrating Goodreads, DBpedia, and Amazon book data into a common schema. The sources contain 52,478, 20,981, and 40,000 records, respectively. A blocking key formed from normalized title and first-author surname yields 10,397 distinct keys in at least two datasets and 94,823 distinct keys in the union. A manual review labels 60 of 60 sampled candidate pairs as matches (20 per dataset pair). These are key-based estimates, not a materialized set of confirmed works. ISBN is reserved for gold-standard evaluation and is not used for matching. The target schema contains `title`, `authors`, `genres`, `publication_year`, `publisher`, `page_count`, `language`, `rating`, `ratings_count`, `price`, `series`, and `awards`; `authors` and `genres` are list-valued.

1 Content

1.1 Use Case

The goal is to integrate book information from three independently maintained sources. They describe overlapping books but use different attribute names, formats, and levels of detail.

The integration maps the sources into a common schema with eight required fields plus `ratings_count`, `price`, `series`, and `awards` (Table 2).

1.2 Requirements Fulfillment

Datasets Three different datasets are used for this project: a Goodreads dataset from Kaggle, book data from DBpedia, and an Amazon Books dataset from Hugging Face. The Goodreads and Amazon datasets are provided in CSV format. The DBpedia data was retrieved in XML format using a SPARQL query executed in Python. The SPARQL query selected all resources of type `dbo:Book` and returned 20,981 results. The exact number of results may vary slightly between query executions.

Table 1 gives an overview of the three datasets, including their sources, formats, number of entities, and available attributes.

The project requirements are addressed as follows:

- **Three datasets:** Goodreads, DBpedia, and Amazon Books satisfy this requirement (Table 1).
- **At least 2,500 entities:** The union contains 94,823 distinct normalized blocking keys, comfortably above the threshold. This is a key-based estimate, not a materialized integrated dataset.
- **At least 1,000 entities in at least two datasets:** The key-based estimate is 10,397 distinct keys, above the threshold.
- **At least eight attributes:** The planned schema has 12, including `title`, `authors`, `genres`, `publication_year`, `publisher`, `page_count`, `language`, and `rating`.
- **At least five attributes contained in at least two datasets:** Ten planned attributes occur in at least two datasets; seven (including `title`, `authors`, `publisher`, `publication_year`, `page_count`, `language`, and `genres`) map across all three (Table 2). DBpedia is sparse for `genres`, `language`, and `year`; voting is mainly planned for `authors`, `publisher`, and `page_count`, missing in 13.5%, 25.4%, and 20.1% of DBpedia records.
- **List attribute and entity identification:** `authors` and `genres` are list-valued; matching uses normalized `title` and first-author surname. ISBN is excluded from matching and reserved for evaluation.

The datasets also differ in their completeness. Attributes with more than 30% missing values are marked with MV (Missing Values) in Table 1. In the Goodreads dataset, this applies to `series`, `characters`, `edition`, `firstPublishDate`, `awards`, and `setting`. The respective missing-value rates are 55.28%, 73.77%, 90.56%, 40.64%, 79.77%, and 77.94%. In the DBpedia dataset, `genres` (49.1%), `language` (87.6%), and `year` (88.5%) exceed this threshold. In the Amazon dataset, no attribute exceeds 30%; the highest share of missing values is 23.5% (`isbn10`).

All three datasets have been downloaded and are available to the team.

Integrated Schema Table 2 maps equivalent source attributes to canonical names: Goodreads `author` and DBpedia/Amazon `authors` map to `authors`; Goodreads `publishDate`, DBpedia `year`, and the year extracted from Amazon `publication_date` map to `publication_year`, with date precision varying by source.

Source representations also differ: Amazon titles may include bracketed series or edition details (“One Small Sacrifice (Shadows of New York, 1)” vs. Goodreads’ “One Small Sacrifice”) and vary in capitalization. Goodreads stores contributors and roles in one string (e.g., “Terry Brooks (Goodreads Author)”), Amazon stores a serialized author list, and DBpedia has author elements. DBpedia’s “English language” corresponds to “English” in Goodreads and Amazon.

Goodreads and Amazon contain ratings, while DBpedia does not; Goodreads adds series and awards. Amazon provides a subtitle field. It is filled for 92.9% of the records, but it mostly contains binding and release information instead of real subtitles.

Table 1. Overview of the analyzed datasets.

Dataset	Source	Format	# Ent.	# Attr.	List of attributes
Goodreads Best Books Ever	Kaggle ¹	CSV	52,478	23	title, series (MV), author, rating, description, language, isbn, genres, characters (MV), bookFormat, edition (MV), pages, publisher, publishDate, firstPublishDate (MV), awards (MV), numRatings, ratingsByStars, likedPercent, setting (MV), bbeScore, bbeVotes, price
DBpedia	DBpedia ²	XML	20,981	8	authors, genres (MV), isbns, language (MV), pages, publisher, title, year (MV)
Amazon Reviews 2023	Hugging Face ³	CSV	40,000	13 ⁴	title, subtitle, authors, publisher, publication_date, language, pages, genres, average_rating, ratings_count, price, isbn13, isbn10

¹ <https://www.kaggle.com/datasets/arnabchaki/goodreads-best-books-ever?resource=download>

² <https://dbpedia.org/sparql>

³ <https://huggingface.co/datasets/McAuley-Lab/Amazon-Reviews-2023>

⁴ Amazon’s 13 attributes exclude the internal identifiers `amazon_id` and `asin`.

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Type	Datasets in which the attribute is found
title	string	GR (title), DB (title), AM (title)
authors	list<string>	GR (author), DB (authors), AM (authors)
genres	list<string>	GR (genres), DB (genres), AM (genres)
publication_year	integer	GR (publishDate), DB (year), AM (publication_date)
publisher	string	GR (publisher), DB (publisher), AM (publisher)
page_count	integer	GR (pages), DB (pages), AM (pages)
language	string	GR (language), DB (language), AM (language)
rating	float	GR (rating), AM (average_rating)
ratings_count	integer	GR (numRatings), AM (ratings_count)
price	float	GR (price), AM (price)
series	string	GR (series)
awards	list<string>	GR (awards)
<i>Identifier for the gold standard only (not used for matching):</i>		
isbn	string	GR (isbn), DB (isbns), AM (isbn13, isbn10)

Notes: Overlap is estimated by normalized matching keys; editions are not merged. GR = Goodreads Best Books Ever, DB = DBpedia, AM = Amazon; source column names in brackets.

Entity Matching and Overlap Candidate works are blocked by normalized title and first-author surname. Normalization lowercases titles, removes diacritics and punctuation, drops bracketed text and (when present) subtitles, and ignores common leading articles. ISBN does not participate in this key; it is reserved for the evaluation gold standard. The overlap script reports 6,562 Amazon–Goodreads, 2,129 Amazon–DBpedia, and 4,634 Goodreads–DBpedia shared keys, with 1,464 keys in all three sources. In total, 10,397 keys occur in at least two sources and the union contains 94,823 distinct keys. A manually labelled sample of 20 candidate pairs per source pair contains 60/60 accepted matches (100% sampled precision). These are key-based estimates, not confirmed counts in a materialized integrated database; title normalization can conflate distinct works, and the sample does not measure recall.

Table 3. Overlap summary by distinct normalized blocking key.

Measure	Count
Amazon $\cap$ Goodreads	6,562
Amazon $\cap$ DBpedia	2,129
Goodreads $\cap$ DBpedia	4,634
All three datasets	1,464
Keys in at least two datasets	10,397
Union of distinct normalized keys	94,823

Table 3 counts distinct keys based on normalized title and first-author surname, so it differs from the record counts in Table 1.